Magic Gems: A Polyhedral Framework for Magic Squares

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Abstract

To an $n \times n$ magic square S we attach the points $\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij}) \in \mathbb{R}^3$, where (x_{ij}, y_{ij}) is the centred cell and z_{ij} the centred entry, and call their convex hull the *Magic Gem* $\mathcal{G}(S)$; its lower and upper faces project onto the regular subdivisions of the $n \times n$ grid induced by the heights z and $-z$. For every semi-magic square of every order the covariance matrix of the point set is $\mathrm{diag}(a_n, a_n, b_n)$ with $a_n \neq b_n$; we deduce that the point set has a non-trivial isometry exactly when the square is self-complementary, and that the point sets of two semi-magic squares are congruent exactly when the squares agree up to the dihedral group and complementation. For the 880 magic squares of order four we compute the gems exactly and show that no gem has an isometry beyond those of its point set: 48 gems are centrally symmetric, 304 have a single half-turn, the other 528 are asymmetric and chiral, and the gems fall into 616 congruence classes, 352 combinatorial types and 39 f-vectors, with 8 to 16 vertices. For the Siamese (de la Loubère) square of every odd order $n \geq 5$ we prove that the gem has no vertices other than the twelve that the boundary of the grid forces (the four corners and, on each side, the cells of the smallest and largest entry), that the gem has f-vector $(12, 26, 16)$, and that its isometry group is $\{\pm\mathrm{id}\}$; the proof reduces to a table of slack polynomials in $m = (n-1)/2$. Over all linear magic squares of orders 5, 7, 9, 11 the largest vertex count is 18, 25, 31, 36, so bounded vertex counts are not a feature of linear constructions in general.

Keywords: magic squares, convex polytopes, regular subdivisions, symmetry groups, Siamese method
MSC 2020: 05B15, 52B05, 52B15, 52B20

1 Introduction

Magic squares are among the oldest objects of recreational mathematics: the 3×3 Lo Shu belongs to Chinese tradition, a 4×4 square is engraved in Dürer's *Melencolia I* (1514), and constructions for every order $n \geq 3$ have been known for centuries [2, 23]. The modern literature treats them from several directions: enumeration and the geometry of the polytopes whose lattice points are the squares [29, 11, 1, 5, 4], spectral and matrix properties [18, 20, 21, 6], and, in work of Loly and collaborators, physical invariants such as the moment of inertia and the multipole moments [16, 25, 26].

Definition 1.1 (Magic square)**.** An *arrangement* of order n is a bijection $S = (s_{ij})$ from the cells of the $n \times n$ grid onto $\{1, \ldots, n^2\}$. It is a *magic square* if every row, every column and both main diagonals sum to the magic constant $\mu_n = n(n^2+1)/2$, and *semi-magic* if only the rows and columns do.

The construction. Centre the grid at the origin and attach to the cell (i, j) the point

$$\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij}),$$

whose horizontal part (x_{ij}, y_{ij}) is the position of the cell and whose height is the *centred entry* $z_{ij} = s_{ij} - \frac{n^2+1}{2}$. Read as vectors from the origin, the $\mathbf{p}_{ij}$ of a magic square have vertical components

that cancel along every row, column and both diagonals (Lemma 2.2; Figures 1 and 2). The *Magic Gem* $\mathcal{G}(S)$ is the convex hull of the n^2 points (Section 2). Figure 1 shows Dürer's square, its vectors and its gem, which has 12 vertices and 4 points in its interior.

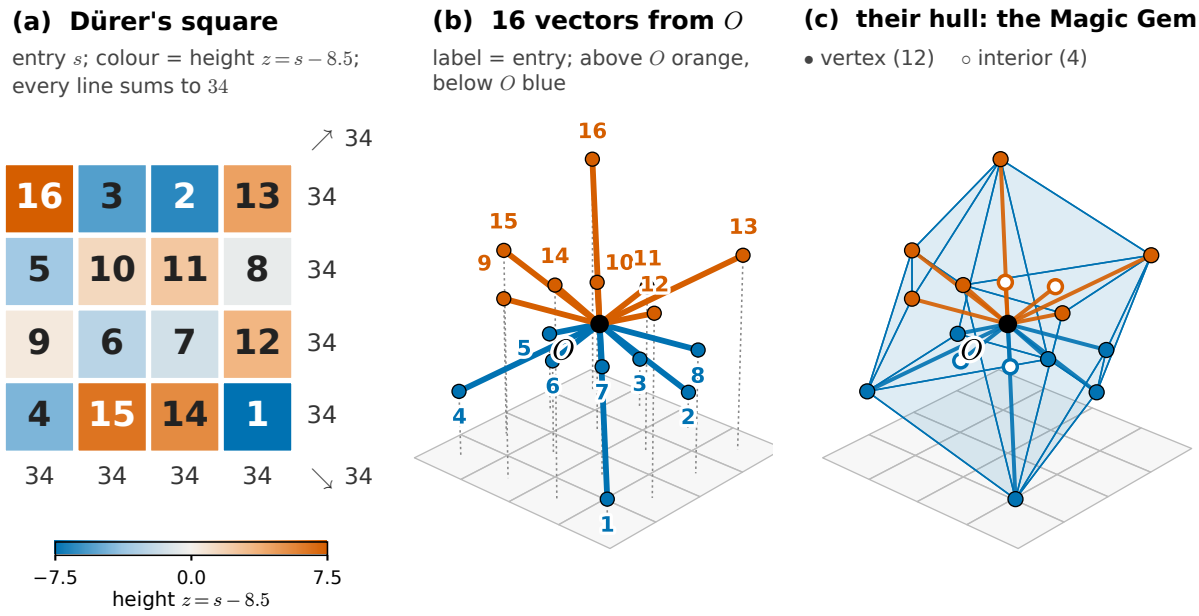


Figure 1: Dürer's square. (a) Entries s_{ij} on a heat map of the heights $z_{ij} = s_{ij} - 8.5$ (orange above 0, blue below); every line sums to 34. (b) The 16 vectors $\mathbf{p}_{ij}$ from the origin O, labelled by the entries, with drop lines to the floor grid. (c) Their convex hull, the Magic Gem: 12 tips are vertices (filled), 4 lie in the interior (open). The colour bar is shared by (a) and (b).

Results. The gem is the lift of the grid by the height function z, so its lower and upper faces are the regular subdivisions of the grid induced by z and $-z$ (Section 2.1).

1. *Symmetry and congruence for every order* (Theorem 4.2). For a semi-magic square the covariance matrix of the $N = n^2$ points is $\mathrm{diag}(a_n, a_n, b_n)$ with $a_n = (n^2-1)/12 \neq b_n = (n^4-1)/12$. Hence every isometry of the point set lies in $D_4 \times \{\pm 1\}$, the point set has a non-trivial isometry if and only if S is self-complementary, and two semi-magic squares have congruent point sets if and only if one is a D_4-image of the other or of its complement.

2. *The* 880 *gems of order four* (Theorem 4.3, Section 5). The isometry group of each gem equals that of its point set; so 48 gems (the associative squares) are centrally symmetric, 304 have one half-turn, and 528 have no symmetry. We tabulate the exact geometry (8 to 16 vertices, 39 f-vectors, 352 combinatorial types, 155 distinct volumes, 616 congruence classes) and show that the vertex set alone determines the square up to D_4 (Proposition 5.1).

3. *Siamese gems* (Theorem 6.2). For the Siamese square of every odd order $n \geq 5$ the vertices of the gem are exactly its twelve forced vertices (Lemma 6.1), the f-vector is $(12, 26, 16)$, and the isometry group is $\{\pm \mathrm{id}\}$. The proof is printed in full as a table of slack polynomials. Over the wider family of linear squares the largest vertex count grows with n (Remark 6.3).

Relation to previous work. The polyhedral geometry of magic squares is classical: nonnegative integer matrices with all line sums equal to t are the lattice points of the t-th dilate of the Birkhoff polytope, and their number is a polynomial in t [29, 11, 31]; the count of magic squares with distinct entries is an Ehrhart quasi-polynomial of an inside-out polytope [5, 4]; the cones of magic squares and cubes have Hilbert-basis descriptions [1]; contingency tables with fixed margins are the same lattice points seen as a sample space [9]. In all of these the magic squares are the *points* of a polytope, whereas the Magic Gem attaches a polytope to a *single* square. The lifting construction

itself is that of regular subdivisions and secondary fans [14, 8]. The moment identities of Section 3 are due in substance to Loly and Rogers [16, 25, 26]; we record them in the coordinates of the gem for later use. The partition of the 880 order-four squares into associative, self-complementary and other squares is Dudeney's [10]. New are the symmetry and congruence theorem for point sets of every order, the hull-level symmetry classification and census of the order-four gems, and the Siamese theorem. The reading of the line conditions as orthogonality conditions on the height function and the statistics of random arrangements are treated in a companion manuscript [19]; no result of it is used here.

Section 2 defines the gem and its relation to regular subdivisions, Section 3 records the moment identities, Section 4 treats symmetry and congruence, Section 5 the order-four census, Section 6 the Siamese gems, and Section 7 lists open questions; Section A describes the computations. An interactive display of the 880 order-four gems is available at `https://kylemath.github.io/MagicGemWebpage/`.

2 The Magic Gem

Let $S = (s_{ij})$ be an arrangement of order $n \geq 2$, with cells indexed by $i, j \in \{0, \dots, n-1\}$ (i the row from the top, j the column from the left), and write $N = n^2$ and $\bar{s} = (N+1)/2$ for the mean entry.

Definition 2.1 (Gem coordinates and Magic Gem)**.** To the cell (i, j) assign the point $\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij}) \in \mathbb{R}^3$ with

$$x_{ij} = j - \frac{n-1}{2}, \qquad y_{ij} = \frac{n-1}{2} - i, \qquad z_{ij} = s_{ij} - \bar{s}. \tag{1}$$

The *Magic Gem* of S is the convex hull $\mathcal{G}(S) = \operatorname{conv}\{\mathbf{p}_{ij}\}$, and $z = (z_{ij})$ is the *height function* of S.

The x-axis runs along the columns and the y-axis along the rows (top row at positive y). Let $x_j = j - \frac{n-1}{2}$ and $y_i = \frac{n-1}{2} - i$ denote the n values taken by x and y. A *line* is one of the n rows, the n columns, the main diagonal $\{i = j\}$ or the anti-diagonal $\{i + j = n-1\}$. For a line L let $\delta_L(S) = \sum_{c \in L} s_c - \mu_n$ be its *deviation*, so S is magic iff $\delta_L(S) = 0$ for all $2n+2$ lines. Moment and centroid statements below refer to the N points $\mathbf{p}_{ij}$ with unit weights, not to the solid $\mathcal{G}(S)$.

Terminology. Each point $\mathbf{p}_{ij}$ is either a *vertex* of $\mathcal{G}(S)$, an *interior point* (in the interior of $\mathcal{G}(S)$), or a *boundary non-vertex point* (on the boundary but not a vertex). We write $\operatorname{vert}\mathcal{G}(S)$ for the vertex set, V, V°, V^∂ for the three numbers ($V + V^\circ + V^\partial = N$), V/N for the *vertex fraction*, and (V, E, F) for the f-vector. Facets are denoted Γ. For any set $X \subseteq \mathbb{R}^3$, $\operatorname{Iso}(X)$ is the group of isometries of $\mathbb{R}^3$ that map X onto itself. The dihedral group $D_4 = \langle r, h \rangle$ of the square acts on arrangements by $(gS)_{gc} = s_c$; here r is the quarter-turn, r^2 the half-turn, h and v the left–right and top–bottom reflections and t the transpose. For $g \in D_4$ let $\hat{g}(x, y, z) = (g(x, y), z)$, where g acts on the plane as the corresponding orthogonal map. The order-four squares are numbered $\#0, \dots, \#879$ in the lexicographic order of their Frénicle standard forms (the image under D_4 with the smallest corner entry at top left and $s_{01} < s_{10}$) [12]; the list is `anc/squares880.txt`.

Lemma 2.2 (Means, variances, line sums)**.** *For every arrangement S of order n:*

1. $\sum_{i,j} x_{ij} = \sum_{i,j} y_{ij} = \sum_{i,j} z_{ij} = 0$ *and* $\sum_{i,j} x_{ij} y_{ij} = 0$;
2. $\frac{1}{N}\sum x_{ij}^2 = \frac{1}{N}\sum y_{ij}^2 = a_n := \frac{n^2-1}{12}$ *and* $\frac{1}{N}\sum z_{ij}^2 = b_n := \frac{N^2-1}{12}$;
3. *for every line L, $\delta_L(S) = \sum_{c \in L} z_c$, and $\sum_{c \in L} \mathbf{p}_c = \big(n\,\bar{x}_L,\ n\,\bar{y}_L,\ \delta_L(S)\big)$, where $(\bar{x}_L, \bar{y}_L)$ is the centre of the line: $(0, y_i)$ for the row i, $(x_j, 0)$ for the column j, and $(0, 0)$ for both diagonals.*

In particular S is magic if and only if the vector sum of the $\mathbf{p}_c$ along every line is horizontal.

Proof. (1) $\sum_{i,j} x_{ij} = n\sum_j x_j = 0$, likewise for y; $\sum z = \sum_{k=1}^{N} k - N\bar{s} = 0$; and $\sum xy = (\sum_j x_j)(\sum_i y_i) = 0$. (2) $\frac{1}{n}\sum_j x_j^2$ is the variance of n equally spaced points with unit spacing, $(n^2-1)/12$; the same holds for y, and for z with N points. (3) Since $\mu_n = n\bar{s}$, $\sum_{c\in L} z_c = \delta_L$. The cells of row i have $y = y_i$ and x running over all x_j; for a column the roles are exchanged; each diagonal is invariant under $(x, y) \mapsto (-x, -y)$. □

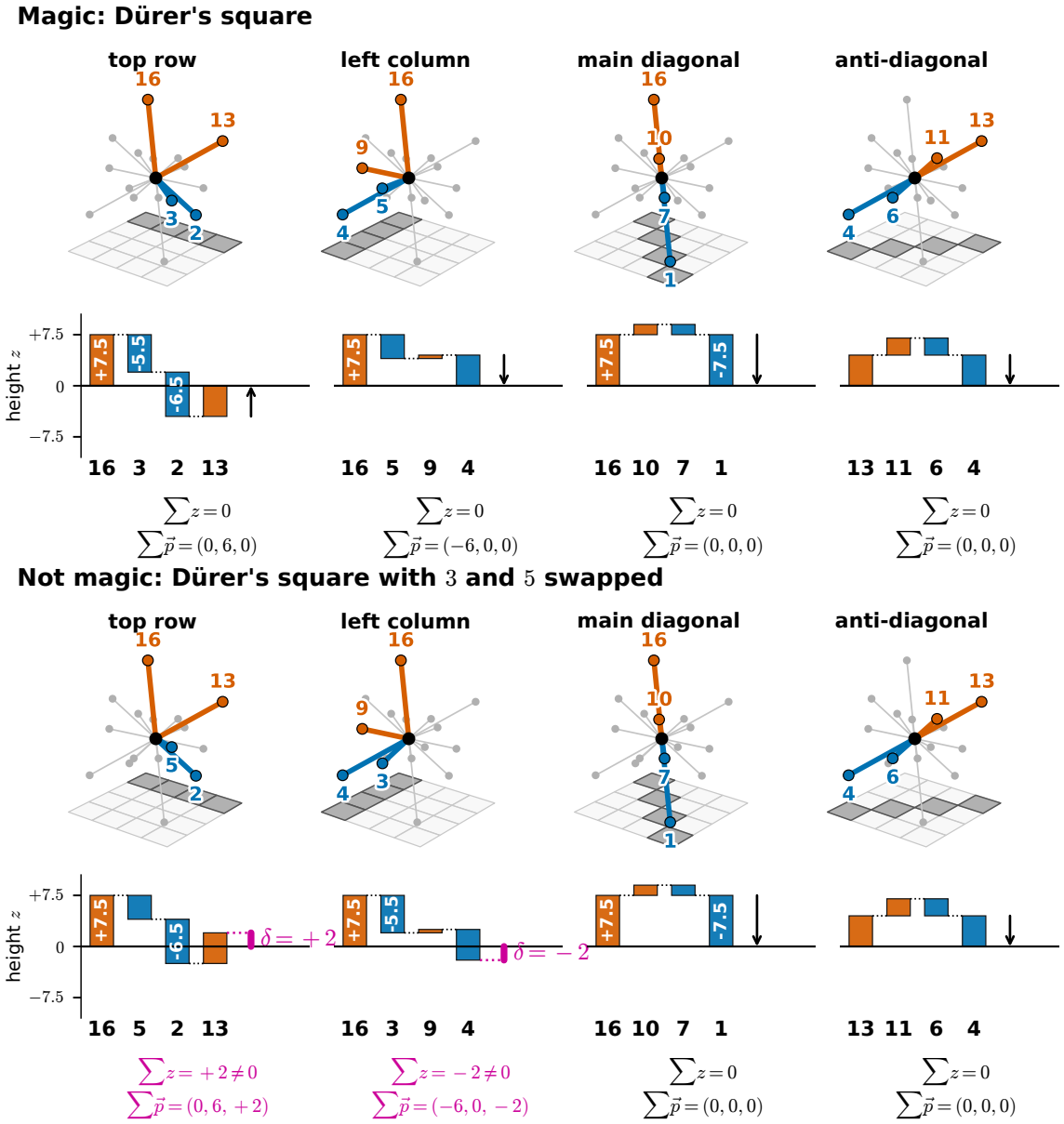


Figure 2: Heights cancel along every line. *Top:* for Dürer's square, the vectors of the top row, left column and both diagonals (coloured; the cells of the line shaded on the floor), and below each the heights $z = s - 8.5$ of the line stacked head to tail; the running total returns to 0, so $\sum_{c\in L} \mathbf{p}_c$ is horizontal (Lemma 2.2(3)). *Bottom:* after swapping the entries 3 and 5 the top row and the left column no longer close (deviations $\delta_L = \pm 2$, magenta); the diagonals do not contain the swapped cells.

For $g \in D_4$ the points of gS are $\hat{g}\,\mathbf{p}_c$, so

$$\mathcal{G}(gS) = \hat{g}\,\mathcal{G}(S), \tag{2}$$

and since D_4 maps magic squares to magic squares, every D_4-class of magic squares has a single gem up to congruence.

Proposition 2.3 (Dimension)**.** *Let S be an arrangement of order $n \geq 2$. The points $\mathbf{p}_{ij}$ are coplanar if and only if S is one of the eight D_4-images of the row-by-row arrangement $s_{ij} = ni + j + 1$. Otherwise $\mathcal{G}(S)$ is a 3-polytope with the origin in its interior. In particular this holds for every semi-magic square.*

Proof. The grid positions are not collinear, so a plane containing all points is not vertical and the points are coplanar if and only if $z = \alpha x + \beta y + \gamma$ on the grid. Then $u = \alpha$ and $w = -\beta$ are the differences of the entries of horizontally and vertically adjacent cells, so they are integers, and $s_{ij} = s_{00} + uj + wi$. Since the entries are distinct, $u, w \neq 0$; applying h, v and t we may assume $1 \leq u \leq w$, so that $s_{00} = 1$ is the smallest entry. The entry 2 is $1 + uj + wi$ with $(i, j) \neq (0, 0)$, hence $u = 1$, and row 0 holds $1, \ldots, n$. The entry $n+1$ lies in a row $i \geq 1$, so $w \leq n$; if $w < n$ then the cells $(0, w)$ and $(1, 0)$ both hold $1 + w$. Hence $w = n$ and $s_{ij} = ni + j + 1$; conversely, every D_4-image of this arrangement has affine heights. If the points span $\mathbb{R}^3$ affinely, their centroid, which is $\mathbf{0}$ by Lemma 2.2(1), is a convex combination with positive weights of all of them and hence an interior point of the hull. Finally, if S is semi-magic and $z = \alpha x + \beta y + \gamma$, then $\gamma = 0$ by Lemma 2.2(1), the row i has height sum $n\beta y_i = 0$ for all i and the column j has $n\alpha x_j = 0$, so $\alpha = \beta = 0$, which is impossible. □

The row-by-row arrangement of order 3 shows that the hypothesis is needed: its gem is a parallelogram.

Example 2.4 (Lo Shu)**.** For the Lo Shu square $S = \begin{pmatrix} 2 & 7 & 6 \\ 9 & 5 & 1 \\ 4 & 3 & 8 \end{pmatrix}$, $\bar{s} = 5$ and the nine points are

$$\begin{array}{lll} \mathbf{p}_{00} = (-1, 1, -3), & \mathbf{p}_{01} = (0, 1, 2), & \mathbf{p}_{02} = (1, 1, 1), \\ \mathbf{p}_{10} = (-1, 0, 4), & \mathbf{p}_{11} = (0, 0, 0), & \mathbf{p}_{12} = (1, 0, -4), \\ \mathbf{p}_{20} = (-1, -1, -1), & \mathbf{p}_{21} = (0, -1, -2), & \mathbf{p}_{22} = (1, -1, 3). \end{array}$$

The origin is the only non-vertex point and is interior; the gem has f-vector $(8, 16, 10)$ and volume 18 (Figure 3).

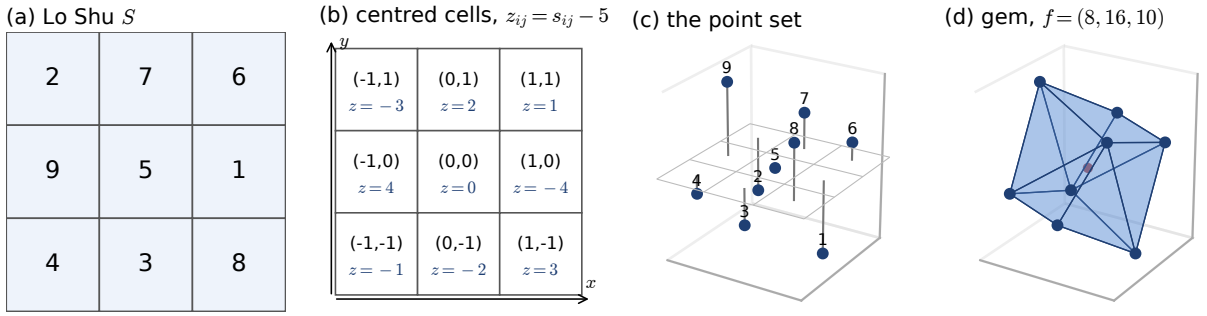


Figure 3: The gem of the Lo Shu square: (a) the square; (b) the centred cells (x_{ij}, y_{ij}) with the heights $z_{ij} = s_{ij} - 5$; (c) the nine points, labelled by the entries; (d) their convex hull, with f-vector $(8, 16, 10)$ and the interior point $\mathbf{0}$.

2.1 Gems and regular subdivisions

Let A be the $n \times n$ grid of cell positions. For a height function $\omega \colon A \to \mathbb{R}$, the *regular subdivision* $\mathcal{S}(\omega)$ of A is obtained by projecting to the plane the lower faces of $\operatorname{conv}\{(c, \omega(c))\}$, i.e. the faces whose outer normals have negative last coordinate; its cells are the sets of grid points whose lifts lie on a common lower face [8, Ch. 2], [14, Ch. 7]. $\mathcal{S}(\omega)$ does not change if ω is multiplied by a positive scalar or an affine function of (x, y) is added to it; the height functions inducing a given

subdivision form a relatively open polyhedral cone, its *secondary cone*. By definition, $\mathcal{G}(S)$ is the lift of A by the height function z of S, and its upper faces are the reflections in $z = 0$ of the lower faces of the lift by $-z$.

Proposition 2.5 (Gems as pairs of regular subdivisions)**.** *Let S be an arrangement of order $n \geq 2$ that is not a D_4-image of the row-by-row arrangement.*

1. *Every facet of $\mathcal{G}(S)$ is a lower facet, an upper facet or a vertical facet. Projection to the plane maps the lower facets onto the maximal cells of $\mathcal{S}(z)$ and the upper facets onto the maximal cells of $\mathcal{S}(-z)$.*

2. *The vertical facets lie in the four planes $x = \pm\frac{n-1}{2}$, $y = \pm\frac{n-1}{2}$; the plane over a side of the grid contains a facet if and only if the entries along that side, in cell order, are not an arithmetic progression.*

3. *$\mathbf{p}_c$ is a vertex of $\mathcal{G}(S)$ if and only if c is a vertex of a cell of $\mathcal{S}(z)$ or of $\mathcal{S}(-z)$.*

4. *The face lattice of $\mathcal{G}(S)$, with vertices labelled by cells, is determined by the pair $(\mathcal{S}(z), \mathcal{S}(-z))$, hence by the secondary cones containing z and $-z$. The complement $N + 1 - S$ exchanges the two subdivisions.*

Proof. (1) is the definition of $\mathcal{S}(\pm z)$. (2) A vertical facet lies in a plane $\ell(x, y) = \max_A \ell$ for a linear form ℓ; the maximisers of ℓ on the grid are a corner or the n cells of a side, and the face is 2-dimensional only in the second case and only if the lifts of the side are not collinear, i.e. if the heights along the side are not an affine function of the position. (3) The vertical line through a vertex $\mathbf{v}$ meets $\mathcal{G}(S)$ in a segment with endpoint $\mathbf{v}$, so $\mathbf{v}$ lies on a lower or an upper face, of which it is a vertex; conversely the vertices of a face are vertices of the polytope. (4) The lower and upper faces together with their incidences are read off from the two subdivisions, and by (2) the vertical facets are the polygons over the sides on which the lower and upper boundary chains differ. The complement has height function $-z$. □

In this language the corner points of the grid, which are vertices of $\operatorname{conv} A$, are vertices of every gem, lying on both the lower and the upper hull, and the other forced vertices of Lemma 6.1 are the lowest and the highest lifted point over each edge of $\operatorname{conv} A$, which are vertices of the restrictions of $\mathcal{S}(z)$ and $\mathcal{S}(-z)$ to that edge. For the 880 order-four gems, 856 have four vertical facets, 20 have three and 4 have two; the height functions of the 7040 order-four magic squares induce 417 distinct regular subdivisions of the 4×4 grid up to D_4, and for 548 of the 880 classes $\mathcal{S}(z)$ is a triangulation (`anc/verify_revision.py`). The census of combinatorial types in Section 5 thus records which pairs of secondary cones of the 4×4 grid contain the pairs $(z, -z)$ for height functions z of magic squares, and the Siamese theorem of Section 6 is the computation of $\mathcal{S}(\pm z)$ for one infinite family, made possible by the invariance of $\mathcal{S}$ under adding affine functions.

3 Moments of the point set

By Lemma 2.2(1) the origin is the centroid of the N points for every arrangement, so the unweighted balance of the vectors $\mathbf{p}_{ij}$ carries no information about S. The line conditions are statements about moments weighted by the heights. Write $R_i = \sum_j z_{ij}$ and $C_j = \sum_i z_{ij}$ for the row and column deviations; S is semi-magic if and only if all R_i and C_j vanish.

Row and column sums as moment conditions. For every function $\varphi\colon \mathbb{R} \to \mathbb{R}$, grouping the cells by column and by row gives

$$\sum_{i,j} z_{ij}\,\varphi(x_{ij}) = \sum_j \varphi(x_j)\,C_j, \qquad \sum_{i,j} z_{ij}\,\varphi(y_{ij}) = \sum_i \varphi(y_i)\,R_i. \tag{3}$$

Since the functions $\varphi(x)$ on the grid span the indicator functions of the columns, S is semi-magic if and only if both sums vanish for every φ. For $\varphi(t) = t$ this says that the *dipole moment* $\sum z\,(x, y)$ of the signed masses z_{ij} (total mass 0) vanishes, in the multipole language of Rogers and Loly [26]; for $\varphi(t) = t^2$ it gives Loly's invariant below. The diagonal conditions are not of this form. The mixed moment $\sum z\,xy$ is not forced: over the 880 order-four magic squares, $8\sum z\,xy$ takes 48 distinct absolute values between 0 and 480 and vanishes for 56 classes (Figure 4b, Figure 5). It vanishes whenever S is associative, since then z is odd under $(x, y) \mapsto (-x, -y)$; this accounts for 48 of the 56.

Corollary 3.1 (Loly [16]). *For every arrangement,* $J(S) := \sum_{i,j} s_{ij}(x_{ij}^2 + y_{ij}^2) = \sum_j x_j^2 C_j + \sum_i y_i^2 R_i + \frac{n^2(n^4-1)}{12}$*; in particular* $J(S) = n^2(n^4 - 1)/12$ *for every semi-magic square.*

Proof. Substitute $s_{ij} = z_{ij} + \bar{s}$, apply (3) with $\varphi(t) = t^2$ to the first part and Lemma 2.2(2) to $\bar{s}\sum(x^2 + y^2) = \bar{s} \cdot 2Na_n$. □

Lemma 3.2 (Covariance). *The covariance matrix* $\Sigma = \frac{1}{N}\sum_{i,j} \mathbf{p}_{ij}\mathbf{p}_{ij}^\top$ *of the points of an arrangement* S *has* $\Sigma_{xx} = \Sigma_{yy} = a_n$, $\Sigma_{zz} = b_n$, $\Sigma_{xy} = 0$, $N\Sigma_{xz} = \sum_j x_j C_j$ *and* $N\Sigma_{yz} = \sum_i y_i R_i$. *If* S *is semi-magic,* $\Sigma = \mathrm{diag}(a_n, a_n, b_n)$, *and* $a_n < b_n$ *for* $n \geq 2$.

Proof. Lemma 2.2(1)–(2) and (3) with $\varphi(t) = t$. □

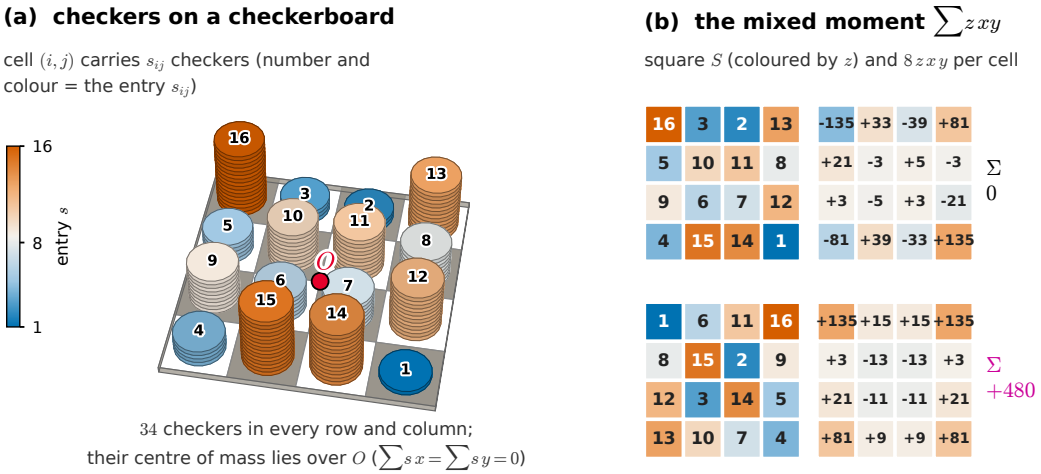


Figure 4: (a) Dürer's square as piles of checkers, s_{ij} on cell (i, j), seen from the front of the board in the orientation of the square; each row and column carries 34 checkers, so their centre of mass lies over the centre O. (b) The mixed moment: for Dürer's square and for square #45 the integers $8\,z\,x\,y$ per cell; they sum to 0 and to 480.

4 Symmetry and congruence

Let $\tau(x, y, z) = (x, y, -z)$ be the reflection in the plane $z = 0$. The *complement* of an arrangement is $\overline{S} = N + 1 - S$, with height function $-z$; S is *self-complementary* if $\overline{S} = gS$ for some $g \in D_4$, and *associative* if $\overline{S} = r^2 S$, i.e. if entries in cells symmetric about the centre add up to $N + 1$. Complements of magic squares are magic. Since the points of $\overline{S}$ are $\tau\mathbf{p}_c$,

$$\mathcal{G}(\overline{S}) = \tau\,\mathcal{G}(S). \tag{4}$$

Lemma 4.1 (Elements of D_4 that can complement). *Let* $\overline{S} = gS$ *for an arrangement* S *of order* $n \geq 2$ *and some* $g \in D_4$. *Then* g *is unique,* $g^2 = \mathrm{id}$, *and* g *fixes no cell except possibly the centre cell of an odd grid, whose entry is then* $\bar{s}$. *Consequently* $g = r^2$ *(so* S *is associative) if* n *is odd, and* $g \in \{r^2, h, v\}$ *if* n *is even. The map* $\iota_g = \hat{g}\,\tau$ *is an isometry of* $\mathbb{R}^3$ *that maps the point set of* S *onto itself, so* $\iota_g \in \mathrm{Iso}(\mathcal{G}(S))$; *explicitly* $\iota_{r^2} = -\mathrm{id}$ *(a point reflection),* $\iota_h(x, y, z) = (-x, y, -z)$ *and* $\iota_v(x, y, z) = (x, -y, -z)$ *(half-turns about the* y- *and the* x-*axis).*

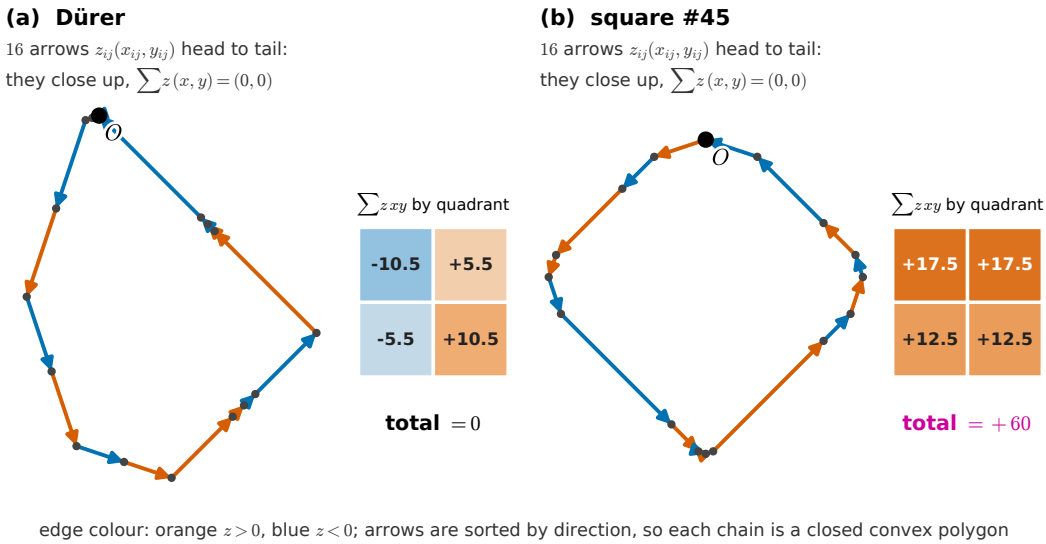


Figure 5: The dipole moment $\sum z_{ij}\,(x_{ij}, y_{ij})$ for Dürer's square and square #45 (Figure 4b): the 16 vectors $z_{ij}(x_{ij}, y_{ij})$ (orange for $z > 0$, blue for $z < 0$), sorted by direction and laid head to tail, close up at O. The tables give $\sum z\,xy$ over the four quadrants of the grid; the totals are 0 and 60.

Proof. If also $\overline{S} = g'S$ then $g^{-1}g'$ fixes S; as the entries are distinct it fixes every cell, and D_4 acts faithfully on the cells, so $g = g'$. Since complementation commutes with g, $g^2S = g\overline{S} = \overline{gS} = \overline{\overline{S}} = S$, so $g^2 = \mathrm{id}$. If g fixes a cell c then $s_c = N + 1 - s_c = 2\bar{s}$. The involutions of D_4 are r^2, h, v and the two diagonal reflections; the diagonal reflections fix $n \geq 2$ cells, h and v fix the n cells of the middle column or row when n is odd and none when n is even, and r^2 fixes only the centre cell when n is odd. Finally, the points of S are mapped by τ to those of $\overline{S} = gS$, and by $\hat{g} = \hat{g}^{-1}$ back to those of $g^2S = S$, by (2). □

Theorem 4.2 (Point sets of semi-magic squares)**.** *Let S and T be arrangements of order $n \geq 2$ whose dipole moments vanish, for instance semi-magic squares, and let P_S, P_T be their point sets.*

1. *Every isometry A of $\mathbb{R}^3$ with $A(P_S) = P_T$ is of the form $\hat{g}$ or $\hat{g}\,\tau$ with $g \in D_4$; $\hat{g}(P_S) = P_T$ if and only if $T = gS$, and $\hat{g}\,\tau(P_S) = P_T$ if and only if $T = g\overline{S}$.*

2. *P_S and P_T are congruent if and only if $T \in D_4S \cup D_4\overline{S}$.*

3. *$\mathrm{Iso}(P_S) = \{\mathrm{id}\}$ unless S is self-complementary, and $\mathrm{Iso}(P_S) = \{\mathrm{id}, \iota_g\}$ if $\overline{S} = gS$.*

Proof. An isometry of $\mathbb{R}^3$ mapping one finite set onto another maps centroid to centroid, and both centroids are $\mathbf{0}$ (Lemma 2.2(1)); so A is orthogonal, and $A\Sigma_S A^\top = \Sigma_T$ for the covariance matrices. By Lemma 3.2, $\Sigma_S = \Sigma_T = \mathrm{diag}(a_n, a_n, b_n)$ with $a_n \neq b_n$, so A commutes with Σ_S and preserves its eigenspaces, the plane $z = 0$ and the z-axis: $A = \mathrm{diag}(B, \varepsilon)$ with $B \in O(2)$ and $\varepsilon = \pm 1$. The projection of both point sets to the plane is the grid, so B is a symmetry of the grid, that is, an element $g \in D_4$; hence $A = \hat{g}$ or $A = \hat{g}\tau$. By (2), $\hat{g}(P_S) = P_{gS}$ and $\hat{g}\tau(P_S) = P_{g\overline{S}}$, and the point set determines the arrangement; this proves (1), and (2) follows. For (3) take $T = S$: $\hat{g}(P_S) = P_S$ forces $gS = S$, hence $g = \mathrm{id}$, and $\hat{g}\tau(P_S) = P_S$ forces $g\overline{S} = S$, i.e. $\overline{S} = g^{-1}S = gS$ by Lemma 4.1. □

So for every order the symmetries of the point set of a semi-magic square are exactly those coming from self-complementarity, and congruences between point sets come only from D_4 and complementation. For the gem, $\mathrm{Iso}(\mathcal{G}(S)) = \mathrm{Iso}(\mathrm{vert}\,\mathcal{G}(S))$, since isometries preserve extreme points, and $\mathrm{Iso}(P_S) \subseteq \mathrm{Iso}(\mathcal{G}(S))$; the inclusion is an equality when every point is a vertex, but in general the gem might have isometries that do not preserve the non-vertex points. For order four this does not happen.

Theorem 4.3 (Isometry groups of the order-four gems; exact computation)**.** *For each of the* 880 *essentially different magic squares* S *of order* 4, $\mathrm{Iso}(\mathcal{G}(S)) = \mathrm{Iso}(P_S)$. *Hence:*

1. *for the* 48 *associative squares,* $\mathrm{Iso}(\mathcal{G}(S)) = \{\mathrm{id}, -\mathrm{id}\}$ *(C_i);*
2. *for the* 304 *self-complementary squares that are not associative,* $\mathrm{Iso}(\mathcal{G}(S)) = \{\mathrm{id}, \iota_g\}$ *with* $g \in \{h, v\}$*, a half-turn about a horizontal coordinate axis (C_2);*
3. *for the remaining* 528 *squares,* $\mathrm{Iso}(\mathcal{G}(S))$ *is trivial.*

In particular a gem has an improper isometry if and only if S *is associative, and the* 832 *gems in cases (2) and (3) are chiral.*

Proof. The squares were sorted into the three cases by comparing $\overline{S}$ with the eight images gS; by Lemma 4.1 the complementing element is r^2, h or v. For each square we computed $\mathrm{Iso}(\mathrm{vert}\,\mathcal{G}(S))$ in exact integer arithmetic (Section A); its order is 2 in cases (1) and (2) and 1 in case (3), equal to the order of $\mathrm{Iso}(P_S)$ by Theorem 4.2(3), and $\mathrm{Iso}(P_S) \subseteq \mathrm{Iso}(\mathcal{G}(S))$. The maps ι_h, ι_v are proper. □

The partition (1)–(3) is classical: the associative squares form Dudeney's group III and the self-complementary non-associative ones his group VI [10]. What Theorem 4.3 adds is that the hulls have no further symmetry. Likewise the congruence classes of the 880 gems are the orbits of complementation (Proposition 5.1). Table 1 summarises the three cases and Figure 6 shows one gem of each.

Table 1: Symmetry types of the 880 order-four gems; "classes" counts congruence classes.

type of S	Dudeney	squares	$\mathrm{Iso}(\mathcal{G})$	classes	V
associative	III	48	$C_i = \{\pm\mathrm{id}\}$	48	$10, 12, 14, 16$
self-complementary, not associative	VI	304	C_2, half-turn	304	$8, 10, 12, 14, 16$
not self-complementary	others	528	trivial	264	$8, \ldots, 16$
total		880		616	

Lemma 4.4 (Parity for associative squares)**.** *Let* S *be associative of order* n *and not a* D_4*-image of the row-by-row arrangement (for instance, associative and semi-magic). Then* V *and* V^∂ *are even and* $V^\circ \equiv n \pmod 2$.

Proof. The point set is invariant under $-\mathrm{id}$ (Lemma 4.1), which therefore permutes the vertices, the interior points and the boundary non-vertex points of the 3-polytope $\mathcal{G}(S)$ (Proposition 2.3). Its only fixed point is $\mathbf{0}$, which is a point $\mathbf{p}_c$ exactly when n is odd (the centre cell), and then an interior point. □

The hypothesis is needed: the row-by-row arrangement of order 3 is associative, but its gem is a parallelogram and has no interior points. For $n = 4$ the 48 associative gems have $V^\circ \in \{0, 2, 4\}$ (4, 23 and 21 gems); among the other squares both parities occur.

Remark 4.5 (Facet orbits). For an associative square, $-\mathrm{id}$ fixes no facet, since no facet plane passes through the interior point $\mathbf{0}$; so F is even. For the half-turn ι_h (the case ι_v is the same after transposition), a facet plane is ι_h-invariant if and only if its normal is parallel to the axis, i.e. if it is one of the vertical planes $y = \pm\frac{n-1}{2}$ over the top and bottom rows; by Proposition 2.5(2) these carry facets unless the row is an arithmetic progression. Among the 304 order-four gems of case (2), 282 have two fixed facets, 18 one and 4 none. Neither isometry fixes a vertex, so V is even for all 352 self-complementary squares.

Remark 4.6 (Central-symmetry defect). For a convex body $K \subset \mathbb{R}^3$ with non-empty interior let $\sigma(K) = \mathrm{vol}\big(\frac{1}{2}(K - K)\big)/\mathrm{vol}(K)$. By the Rogers–Shephard inequalities [27], $1 \le \sigma(K) \le \binom{6}{3}/8 = \frac{5}{2}$, with equality on the left if and only if K is centrally symmetric and on the right if and only if K is a simplex; see [15, 28] for this and other measures of symmetry. By Theorem 4.3, $\sigma = 1$ exactly for the 48 associative gems of order four; for the other 832, an exact computation (`anc/central_symmetry_defect.py`) gives $\sigma \in [1057/1018,\ 145/88] \approx [1.038,\ 1.648]$.

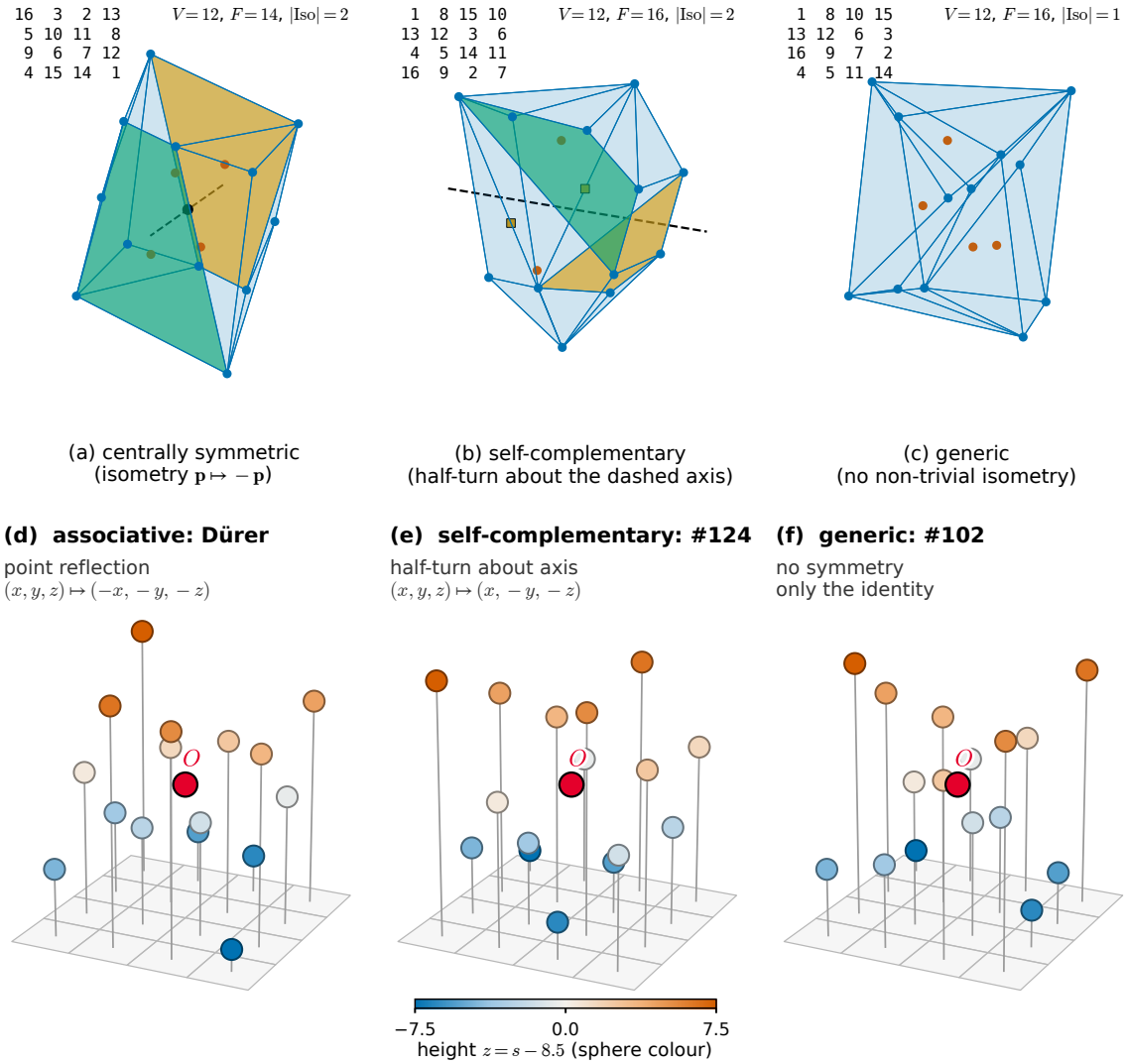


Figure 6: Gems (top) and point sets (bottom) of three order-four squares, all drawn from the same viewpoint, with the square printed above each gem. Interior points: red discs; boundary non-vertex points: orange squares. (a, d) Dürer's square, associative: a pair of opposite facets $\Gamma, -\Gamma$ (orange, green), and the 8 antipodal pairs $\{\mathbf{p}, -\mathbf{p}\}$ joined through O. (b, e) Square #124, self-complementary and not associative: the half-turn about the dashed axis exchanges the highlighted facets and the 8 pairs of points. (c, f) Square #102: no non-trivial isometry; the gem is chiral and its mirror image in $z = 0$ is the gem of $17 - S$.

5 The order-four census

There are 8 magic squares of order 3, all images of the Lo Shu under D_4, and 7,040 of order 4, forming 880 D_4-classes; the order-four squares were listed by Frénicle de Bessy [12] and have been recounted by computer [4]. Dudeney sorted them into twelve groups according to the pattern of complementary pairs $(a, 17 - a)$ [10]; since the numbering of the groups differs between reproductions of his table, we list the labels used here, for all 7040 squares, in `anc/all7040_dudeney.csv`. The number 275,305,224 of essentially different order-five squares is due to Schroeppel (1973) [13]; for order six the count was estimated by Monte Carlo methods [24] and was computed exactly by H. Mino in 2024 [30].

For each of the 880 squares we computed $\mathcal{G}(S)$ exactly (integer facet inequalities, rational volumes; Section A). Table 2 and Figure 7 summarise the results.

Vertices, faces and volumes. Between 8 and all 16 points are vertices; $V = 12$ is the most frequent value (380 gems), and the mean is $269/22 \approx 12.23$. The 39 f-vectors range from $(8, 14, 8)$

Table 2: The 880 order-four gems by number V of vertices: numbers of squares, of combinatorial types (face lattices up to isomorphism), of f-vectors and of distinct exact volumes, and the volume as a fraction of the box $\mathcal{B}_4$ ($\operatorname{vol}\mathcal{B}_4 = 135$). A volume can occur for several values of V, so the volume counts add up to more than 155. Bottom: numbers of gems with k interior points ($V^\circ = k$) and with k boundary non-vertex points ($V^\partial = k$).

V	8	9	10	11	12	13	14	15	16	all
squares	7	10	103	80	380	106	162	14	18	880
types	4	1	27	32	129	52	85	7	15	352
f-vectors	4	1	6	5	7	5	5	3	3	39
volumes	4	5	51	33	111	40	65	6	14	155
min of $\operatorname{vol}/\operatorname{vol}\mathcal{B}_4$	.622	.588	.481	.549	.430	.456	.452	.498	.449	.430
mean of $\operatorname{vol}/\operatorname{vol}\mathcal{B}_4$	.642	.623	.631	.630	.634	.614	.608	.560	.536	.623
max of $\operatorname{vol}/\operatorname{vol}\mathcal{B}_4$	.648	.667	.701	.699	.721	.688	.691	.615	.598	.721

k	0	1	2	3	4	5	6
gems with $V^\circ = k$	37	32	353	178	280	0	0
gems with $V^\partial = k$	357	204	257	40	21	0	1

(one gem) to $(16, 36, 22)$ (nine gems); the most frequent is $(12, 26, 16)$ (164 gems). For these 880 gems the multiset of facet sizes is determined by the f-vector. The volume takes 155 distinct values, from 58 to 292/3, with mean 84.05; its denominators are $1, 2, 3, 6$, because for even n all coordinates are half-odd integers, so that $\mathcal{G}(S) + (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is a lattice polytope. The Ehrhart polynomials of these lattice polytopes take 554 distinct values (`anc/ehrhart_880.py`); for lattice 3-polytopes this polynomial is equivalent to the volume, the lattice-normalised surface area and the number of lattice points. Every gem lies in the box $\mathcal{B}_n = [-\frac{n-1}{2}, \frac{n-1}{2}]^2 \times [-\frac{N-1}{2}, \frac{N-1}{2}]$, and the gems of order four fill between 43% and 72% of it. At most four points are interior, and at most six are boundary non-vertex points.

Combinatorial types. Two gems have the same combinatorial type if their vertex–facet incidence graphs are isomorphic. Grouping by a Weisfeiler–Lehman hash of these graphs and confirming each group by an explicit isomorphism test gives 352 types. By (4) the gems of S and $\overline{S}$ are mirror images, so every type is a union of complement orbits; the 93 types with one member are self-complementary squares, and of the 201 types with two members, 177 are mirror pairs and 24 consist of two self-complementary squares. The numbers of types with $1, 2, 3, \ldots$ members are

squares per type	1	2	3	4	5	6	7	8	10	13	14	16	26	32
types	93	201	8	20	4	7	4	5	5	1	1	1	1	1

The largest type, with $f = (10, 22, 14)$ (twelve triangles and two quadrilaterals), has 32 members. The vertex count is not tied to Dudeney's groups: every group with more than eight members spans at least three vertex counts and at least 20 types, and the 8-vertex gems occur only in groups VI (three) and VIII (four). The 48 pandiagonal squares (Dudeney's group I [22]) have gems with 11 to 15 vertices and between one and four interior points; no order-four square is both pandiagonal and associative.

Proposition 5.1 (Injectivity and congruence)**.** *Let S and T be magic squares of order four.*

1. *If $\hat{g}\operatorname{vert}\mathcal{G}(S) = \operatorname{vert}\mathcal{G}(T)$ for some $g \in D_4$, then $T = gS$. In other words, the vertex set of the gem, up to the action of D_4, determines the D_4-class of the square.*

2. *$\mathcal{G}(S)$ and $\mathcal{G}(T)$ are congruent if and only if $T \in D_4S \cup D_4\overline{S}$. There are* 616 *congruence classes of gems:* 352 *self-complementary classes and* 264 *mirror pairs.*

Proof. (1) The 880 vertex sets have pairwise distinct D_4-canonical forms (exact comparison), so $T = g'S$ for some $g' \in D_4$, and then $\hat{g}'\operatorname{vert}\mathcal{G}(S) = \operatorname{vert}\mathcal{G}(T) = \hat{g}\operatorname{vert}\mathcal{G}(S)$ by (2). Thus $\hat{g}^{-1}\hat{g}'$ is

an isometry of $\mathcal{G}(S)$ that preserves heights; by Theorem 4.3 the only such isometry is the identity, so $g' = g$. (2) "If" is (2) and (4). For "only if" we tested all pairs of vertex sets for a bijection preserving all squared distances, which is equivalent to congruence (Section A); the congruence classes are exactly the 616 complement orbits. □

Forced vertices and named squares. For order four the forced vertices of Lemma 6.1 number between 4 and 13, and nothing more is forced: the only cells that carry a vertex in all 880 gems are the four corners, and the only entries that always label a vertex are 1 and 16. In eight gems the point of the entry 2 is a boundary non-vertex point, for instance for square #593 below. The gems of the Lo Shu, of Dürer's square and of the Siamese square of order 5 have vertex fractions 8/9, 12/16 and 12/25 and volumes 18, 92 and 830/3. The gem of Dürer's square is centrally symmetric with twelve vertices; exchanging its first two rows and its last two rows gives square #802, whose gem has all sixteen points as vertices. The squares cited in Figures 4, 6 and 7 and in the text are

$$\#45 = \begin{pmatrix} 1 & 6 & 11 & 16 \\ 8 & 15 & 2 & 9 \\ 12 & 3 & 14 & 5 \\ 13 & 10 & 7 & 4 \end{pmatrix} \quad \#54 = \begin{pmatrix} 1 & 6 & 11 & 16 \\ 15 & 12 & 5 & 2 \\ 14 & 9 & 8 & 3 \\ 4 & 7 & 10 & 13 \end{pmatrix} \quad \#102 = \begin{pmatrix} 1 & 8 & 10 & 15 \\ 13 & 12 & 6 & 3 \\ 16 & 9 & 7 & 2 \\ 4 & 5 & 11 & 14 \end{pmatrix} \quad \#124 = \begin{pmatrix} 1 & 8 & 15 & 10 \\ 13 & 12 & 3 & 6 \\ 4 & 5 & 14 & 11 \\ 16 & 9 & 2 & 7 \end{pmatrix}$$

$$\#379 = \begin{pmatrix} 2 & 12 & 5 & 15 \\ 13 & 7 & 10 & 4 \\ 16 & 6 & 11 & 1 \\ 3 & 9 & 8 & 14 \end{pmatrix} \quad \#593 = \begin{pmatrix} 4 & 1 & 16 & 13 \\ 5 & 15 & 2 & 12 \\ 14 & 8 & 9 & 3 \\ 11 & 10 & 7 & 6 \end{pmatrix} \quad \#802 = \begin{pmatrix} 5 & 10 & 11 & 8 \\ 16 & 3 & 2 & 13 \\ 4 & 15 & 14 & 1 \\ 9 & 6 & 7 & 12 \end{pmatrix}$$

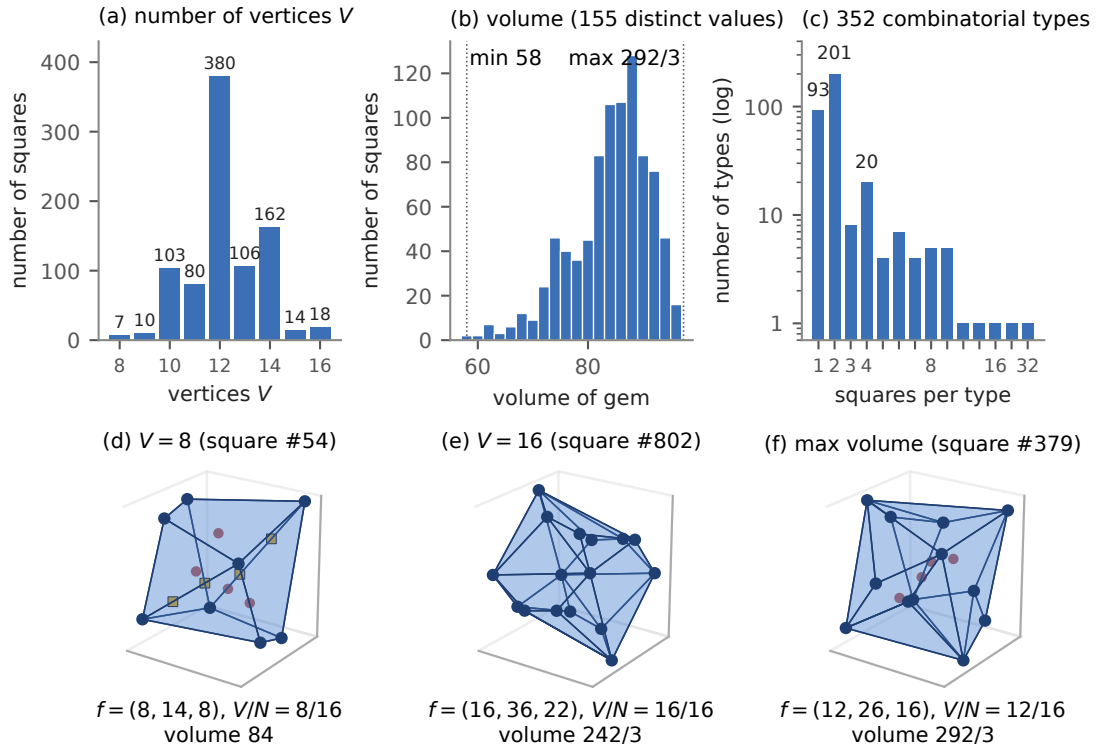


Figure 7: The 880 order-four gems. (a) Number of vertices. (b) Exact volume. (c) Sizes of the 352 combinatorial types (logarithmic scale). (d)–(f) Three extreme gems with translucent faces: the unique gem with $f = (8, 14, 8)$ (square #54; interior points as red discs, boundary non-vertex points as orange squares), a gem with all 16 points as vertices (square #802), and the gem of largest volume (square #379).

6 Siamese gems

6.1 Forced vertices

The *sides* of the grid are its first and last rows and columns.

Lemma 6.1 (Forced vertices)**.** *Let S be an arrangement of order $n \geq 2$. The following points are vertices of $\mathcal{G}(S)$: the points over the four corner cells; the points carrying the entries 1 and N; and, for each side, the points over the cells of the smallest and of the largest entry on that side.*

Proof. A point that is the unique maximiser of a linear functional over the N points is a vertex. The entries are distinct, so $\mp z$ have unique maximisers where $s = 1$ and $s = N$. The functional $-x + y$ attains its maximum only at the top-left corner, and similarly for the other corners. On the left column $x = -\frac{n-1}{2}$, while $x \geq -\frac{n-1}{2} + 1$ elsewhere; for $\varepsilon = 1/N$ the functional $-x + \varepsilon z$ exceeds $\frac{n-1}{2} - \frac{1}{2}$ on the left column and is at most $\frac{n-1}{2} - \frac{1}{2}$ elsewhere, so its unique maximiser is the point of largest height on the left column; $-x - \varepsilon z$ gives the smallest, and the other sides follow by symmetry. □

In the language of Proposition 2.5, the lemma says that the vertices of $\operatorname{conv} A$ and the extreme heights over each edge of $\operatorname{conv} A$ are used by every pair of subdivisions $\mathcal{S}(z), \mathcal{S}(-z)$. For order four the forced vertices number between 4 and 13 and in general other points are vertices as well (Section 5). For the Siamese squares nothing else is.

6.2 The theorem

The *Siamese* (de la Loubère) square of odd order $n = 2m+1$ [7] has 1 in the middle cell of the top row; each next entry is placed one step up and one step to the right, cyclically, or one step down if that cell is occupied. For $n = 5$ it is

$$\begin{pmatrix} 17 & 24 & 1 & 8 & 15 \\ 23 & 5 & 7 & 14 & 16 \\ 4 & 6 & 13 & 20 & 22 \\ 10 & 12 & 19 & 21 & 3 \\ 11 & 18 & 25 & 2 & 9 \end{pmatrix},$$

and for $n = 3$ it is a D_4-image of the Lo Shu. Write $\langle t \rangle \in \{0, \dots, n-1\}$ for the residue of t modulo n.

Theorem 6.2 (Siamese gems)**.** *Let S be the Siamese square of odd order $n = 2m + 1 \geq 3$. The vertices of $\mathcal{G}(S)$ are exactly the forced vertices of Lemma 6.1: the eight boundary points for $n = 3$, and for $n \geq 5$ the twelve points over the cells*

$$\begin{array}{lll} (0,0)\colon \bar{s} + m + 2, & (0, m-1)\colon N - 1, & (0, m)\colon 1, \\ (0, 2m)\colon \bar{s} + m, & (m-1, 0)\colon N - m, & (m, 0)\colon m + 2, \\ (m, 2m)\colon N - m - 1, & (m+1, 2m)\colon m + 1, & (2m, 0)\colon \bar{s} - m, \\ (2m, m)\colon N, & (2m, m+1)\colon 2, & (2m, 2m)\colon \bar{s} - m - 2, \end{array}$$

listed with their entries. For $n \geq 5$ the gem has f-vector $(12, 26, 16)$, with 12 triangular and 4 quadrilateral facets, and $\operatorname{Iso}(\mathcal{G}(S)) = \operatorname{Iso}(P_S) = \{\mathrm{id}, -\mathrm{id}\}$.

So the vertex fraction $12/n^2$ tends to zero, and the combinatorial type of the gem does not depend on $n \geq 5$.

Proof. Step 1: closed form. We show $s(i,j) = n\langle i+j-m \rangle + \langle i+2j-2m \rangle + 1$, the classical description of the Siamese square as a linear square [2]. The map $\kappa(i,j) = (\langle i + j - m \rangle, \langle i + 2j - 2m \rangle)$ is a bijection of $\mathbb{Z}_n^2$ (determinant 1), and $\kappa(0,m) = (0,0)$. Write $k - 1 = nq + r$ with $0 \leq r < n$; we prove by induction on k that the entry k is placed in the cell $\kappa^{-1}(q,r)$. This holds for $k = 1$. Suppose it holds for $1, \dots, k$, with k in the cell $c = \kappa^{-1}(q,r)$; the occupied cells are $\kappa^{-1}(q', r')$ with $nq' + r' < k$. The up-right move $(i,j) \to (i-1, j+1)$ keeps $i+j$ and increases $i+2j$ by 1, so it leads to $\kappa^{-1}(q, \langle r+1 \rangle)$. If $r < n-1$, this cell is empty and is the cell prescribed for $k+1$. If $r = n-1$, i.e. $n \mid k$, it is the cell $\kappa^{-1}(q, 0)$ of the entry $nq + 1$, so the move is blocked; the downward move $(i,j) \to (i+1, j)$ increases both $i + j$ and $i + 2j$ by 1 and leads to the empty cell $\kappa^{-1}(q+1, 0)$,

the cell prescribed for $k+1 = n(q+1)+1$. (Thus each run of n entries fills one broken diagonal $i+j \equiv \text{const.}$)

Step 2: the forced vertices. In every row and every column the leading digit $\langle i+j-m\rangle$ takes each value in $\{0,\dots,n-1\}$ once, so the largest entry of a side is in the cell where this digit is $n-1$ and the smallest where it is 0. On the top row $i=0$ this gives $j=m-1$ with entry $n(n-1)+\langle -2\rangle+1 = N-1$, and $j=m$ with entry 1; on the bottom row $i=2m$, $j=m$ with entry $n(n-1)+n = N$, and $j=m+1$ with entry $\langle n+1\rangle+1 = 2$; on the left column $j=0$, $i=m-1$ with entry $n(n-1)+\langle -m-1\rangle+1 = N-m$, and $i=m$ with $\langle -m\rangle+1 = m+2$; on the right column $j=2m$, $i=m$ with entry $n(n-1)+\langle 3m\rangle+1 = N-m-1$, and $i=m+1$ with $\langle 3m+1\rangle+1 = m+1$. The corner entries follow in the same way. The entries 1 and N are among the side extremes, and for $m\ge 2$ the eight side cells are distinct and are not corners; so the forced vertices are the twelve listed points. For $m=1$ the side extremes are the corners and the midpoints of the sides. It remains to show that no other point is a vertex. For $n=3$ this holds since the only other point is the centre $\mathbf{0}$, an interior point; let $m\ge 2$.

Step 3: affine reduction to five layers. With $x=j-m$, $y=m-i$ and $\langle t\rangle = t-n\lfloor t/n\rfloor$ the closed form gives

$$z = \lambda(x,y)+n\,\zeta(x,y), \qquad \lambda(x,y) = n(x-y+m)+(2x-y+m)+1-\bar{s},$$
$$\zeta(x,y) = -n\left\lfloor \tfrac{x-y+m}{n}\right\rfloor - \left\lfloor \tfrac{2x-y+m}{n}\right\rfloor.$$

The map $\Phi(x,y,z) = (x,y,(z-\lambda(x,y))/n)$ is an affine bijection that preserves vertical lines, so $\Phi(\mathcal{G}(S)) = \operatorname{conv}\{(x,y,\zeta(x,y))\}$ has the same face lattice and vertex cells as $\mathcal{G}(S)$; equivalently, $\mathcal{S}(\pm z) = \mathcal{S}(\pm\zeta)$. For $|x|,|y|\le m$ both floors lie in $\{-1,0,1\}$, and a case check of the two floors shows that ζ takes five values, on the lattice points of five convex lattice polygons in the box $|x|,|y|\le m$ (Figure 8a): $\zeta = n+1$ on $\Lambda_{+2} = \{y\ge x+m+1\} = \operatorname{conv}\{(-m,m),(-1,m),(-m,1)\}$; $\zeta=1$ on $\Lambda_{+1} = \{2x+m+1\le y\le x+m\} = \operatorname{conv}\{(-m,0),(-m,1-m),(-1,m-1)\}$; $\zeta=0$ on $\Lambda_0 = \{|y-2x|\le m\} = \operatorname{conv}\{(-m,-m),(0,m),(m,m),(0,-m)\}$; and $\zeta=-1$, $-(n+1)$ on $\Lambda_{-1}=-\Lambda_{+1}$, $\Lambda_{-2}=-\Lambda_{+2}$.

Step 4: the hull. Let $v_1,\dots,v_{12}$ be the images under Φ of the twelve forced vertices and $u_1,\dots,u_4$ the remaining four vertices of the layer polygons; their coordinates are in Table 3. Let Q be the polyhedron defined by the inequalities

$$\begin{aligned}
&(1)\ \zeta \le 2m+2, \qquad (2)\ y\le m, \qquad (3)\ -x\le m, \qquad (8)\ -2y+\zeta\le 2m,\\
&(4)\ -2(m{+}1)x+(2m{+}1)y-m\zeta\le m(2m{+}1), \qquad (7)\ 2mx-y+m\zeta\le m(2m{-}1),\\
&(5)\ 2(m{+}1)x-2(m{+}1)y+(2m{+}1)\zeta\le 2m(m{+}1),\\
&(6)\ 2m(m{+}2)x-(2m^2{+}2m{-}1)y+m(2m{+}1)\zeta\le m(2m^2{+}2m{-}1),
\end{aligned}$$

and their images (1′)–(8′) under $\mathbf{q}\mapsto -\mathbf{q}$. Table 3 lists the slack (right-hand side minus left-hand side) of (1)–(8) at the sixteen layer vertices. Every entry is 0 or a product of factors that are positive for $m\ge 2$, and the set of layer vertices is centrally symmetric, so all sixteen inequalities hold at all layer vertices and hence, by convexity, at all points (x,y,ζ): $\Phi(\mathcal{G}(S))\subseteq Q$. The zeros of the table show that the planes (1)–(8) contain the following v_k, in cyclic order:

$$\begin{aligned}
&(1)\colon v_1v_2v_3, \quad (2)\colon v_1v_2v_8v_7, \quad (3)\colon v_1v_3v_5v_4, \quad (4)\colon v_1v_4v_7,\\
&(5)\colon v_2v_3v_6, \quad (6)\colon v_2v_6v_9, \quad (7)\colon v_2v_8v_9, \quad (8)\colon v_3v_5v_6,
\end{aligned}$$

with the antipodal lists for (1′)–(8′) ($v_{10}=-v_1$, $v_{11}=-v_3$, $v_{12}=-v_2$, $v_9=-v_4$, $v_8=-v_5$, $v_7=-v_6$). Hence each of the sixteen planes meets $P=\operatorname{conv}\{v_1,\dots,v_{12}\}\subseteq Q$ in a polygon, which is therefore a facet of P. These sixteen polygons (12 triangles and the 4 quadrilaterals $(2),(3),(2'),(3')$, which are convex in the cyclic orders given) have 26 edges, each shared by exactly two of them; since the facet-adjacency graph of a 3-polytope is connected, they are all the facets of P, and $P=Q$. As $P\subseteq\Phi(\mathcal{G}(S))\subseteq Q=P$, the gem is the image of P under Φ^{-1}, with vertex set the twelve forced vertices and f-vector $(12,26,16)$.

Step 5: isometries. S is associative (by the closed form, $s(i,j)+s(2m-i,2m-j)=N+1$), so $\mathrm{Iso}(P_S)=\{\pm\mathrm{id}\}$ by Theorem 4.2. An isometry of $\mathcal{G}(S)$ fixes the centroid $\mathbf{0}$ of its centrally symmetric vertex set and is orthogonal. In the original coordinates $v_1=(-m,m,m+2)$, $v_2=(-1,m,2m^2+2m-1)$, $v_3=(-m,1,2m^2+m)$, $v_4=(-m,0,-2m^2-m+1)$, $v_5=(-m,-m,-m)$, $v_6=(0,-m,2m^2+2m)$, and the other six are their negatives. The squared norms $3m^2<3m^2+4m+4<4m^4+4m^3-2m^2-2m+1<4m^4+4m^3+2m^2+1<4m^4+8m^3+m^2-4m+2<4m^4+8m^3+5m^2$ of v_5,v_1,v_4,v_3,v_2,v_6 are distinct for $m\geq 2$, so an isometry maps each v_k to $\pm v_k$. The inner products of v_1 with $v_2,\dots,v_6$ are $2m^3+7m^2+4m-2$, $m(2m^2+6m+3)$, $-(2m^3+4m^2+m-2)$, $-m(m+2)$, $m(2m^2+5m+4)$, all non-zero, so all signs agree with that of v_1, and the isometry is $\pm\mathrm{id}$. □

Table 3: Proof of Theorem 6.2, Step 4: the sixteen vertices of the layer polygons $\Lambda_{+2},\Lambda_{+1},\Lambda_0,\Lambda_{-1},\Lambda_{-2}$ (groups of rows) and the slacks of the inequalities (1)–(8) at them, as polynomials in m; zeros in bold. All non-zero entries are positive for $m\geq 2$. The points $v_1,\dots,v_{12}$ are the images of the forced vertices, $u_1,\dots,u_4$ are not vertices of the gem.

point	(x,y,ζ)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
v_1	$(-m,m,2m+2)$	**0**	**0**	**0**	**0**	$2(m-1)(m+1)$	$2m(m-1)(m+2)$	$2m(m-1)$	$2(m-1)$
v_2	$(-1,m,2m+2)$	**0**	**0**	$m-1$	$2(m-1)(m+1)$	**0**	**0**	**0**	$2(m-1)$
v_3	$(-m,1,2m+2)$	**0**	$m-1$	**0**	$(m-1)(2m+1)$	**0**	$(m-1)(2m+1)$	$(m-1)(2m-1)$	**0**
v_4	$(-m,0,1)$	$2m+1$	m	**0**	**0**	$4m^2+2m-1$	$2m(2m^2+2m-1)$	$2m(2m-1)$	$2m-1$
u_1	$(-m,1-m,1)$	$2m+1$	$2m-1$	**0**	$(m-1)(2m+1)$	$2m^2+2m+1$	$(m+1)(2m^2+2m-1)$	$4m^2-3m+1$	1
u_2	$(-1,m-1,1)$	$2m+1$	1	$m-1$	$m-1$	$4m^2+2m-1$	$(m+1)(4m^2-2m+1)$	$(m+1)(2m-1)$	$4m-3$
v_5	$(-m,-m,0)$	$2(m+1)$	$2m$	**0**	$2m^2$	$2m(m+1)$	$2m^2(m+2)$	$2m(2m-1)$	**0**
v_6	$(0,-m,0)$	$2(m+1)$	$2m$	m	$2m(2m+1)$	**0**	**0**	$2m(m-1)$	**0**
v_7	$(0,m,0)$	$2(m+1)$	**0**	m	**0**	$4m(m+1)$	$2m(2m^2+2m-1)$	$2m^2$	$4m$
v_8	$(m,m,0)$	$2(m+1)$	**0**	$2m$	$2m(m+1)$	$2m(m+1)$	$2m(m-1)(m+1)$	**0**	$4m$
v_9	$(m,0,-1)$	$2m+3$	m	$2m$	$2m(2m+1)$	$2m+1$	**0**	**0**	$2m+1$
u_3	$(m,m-1,-1)$	$2m+3$	1	$2m$	$(m+1)(2m+1)$	$2m^2+2m-1$	$(m-1)(2m^2+2m-1)$	$m-1$	$4m-1$
u_4	$(1,1-m,-1)$	$2m+3$	$2m-1$	$m+1$	$4m^2+m+1$	$2m+1$	$(m-1)(2m+1)$	$(m-1)(2m-1)$	3
v_{10}	$(m,-m,-2m-2)$	$4(m+1)$	$2m$	$2m$	$2m(2m+1)$	$2(m+1)^2$	$2m(m^2+m+1)$	$2m^2$	$2(m+1)$
v_{11}	$(m,-1,-2m-2)$	$4(m+1)$	$m+1$	$2m$	$(m+1)(2m+1)$	$4m(m+1)$	$(m+1)(4m^2-2m+1)$	$(m+1)(2m-1)$	$4m$
v_{12}	$(1,-m,-2m-2)$	$4(m+1)$	$2m$	$m+1$	$2(m^2+m+1)$	$4m(m+1)$	$2m(2m^2+2m-1)$	$2m(2m-1)$	$2(m+1)$

All entries of Table 3 and the closure of the facet list are verified by `anc/siamese_slacks.py` in exact polynomial arithmetic, and the theorem is checked against exact hull computations for $n=3,5,\dots,41$ (`anc/verify_revision.py`). For $m=1$ several entries vanish and the four triangles degenerate.

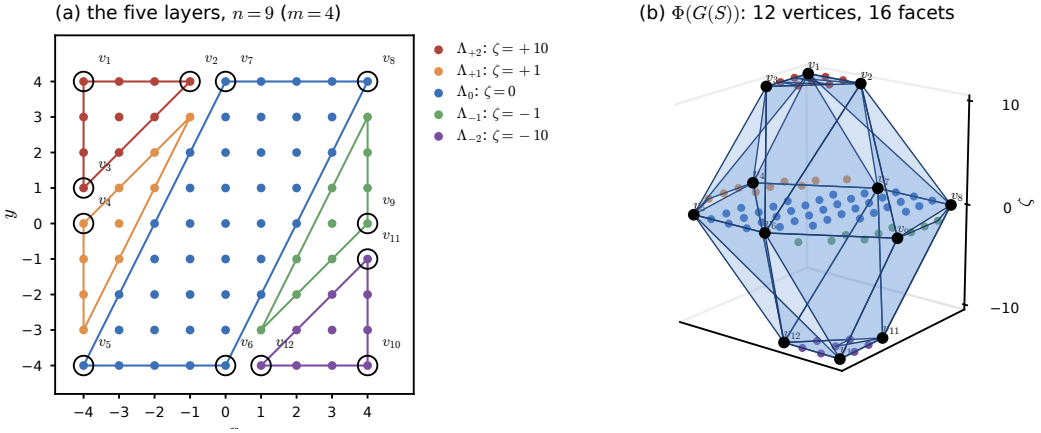


Figure 8: The Siamese gem of order $n=9$ after the affine reduction Φ. (a) The 81 cells coloured by the layer value $\zeta\in\{\pm10,\pm1,0\}$, with the five layer polygons outlined and the twelve vertex cells $v_1,\dots,v_{12}$ circled. (b) $\Phi(\mathcal{G}(S))$ with its 12 vertices and 16 facets. The combinatorics of the layer polygons does not depend on $m\geq 2$.

Remark 6.3 (Linear squares). The Siamese square is a *linear* (cyclic) square $s(c)=n\langle\mathbf{u}\cdot c+\alpha\rangle+\langle\mathbf{w}\cdot c+\beta\rangle+1$ with $\mathbf{u},\mathbf{w}\in\mathbb{Z}_n^2$ and $\alpha,\beta\in\mathbb{Z}_n$, and for fixed integer vectors $\mathbf{u},\mathbf{w}$ the same shear reduces the lift to a number of layers that does not depend on n. The gems of linear squares do not share the bounded vertex count, however. Over all linear magic squares of order n (`anc/linear_squares.py`,

exact hulls) the vertex counts range over $[11, 18]$ for $n = 5$ (184 D_4-classes), $[12, 25]$ for $n = 7$ (3159 classes), $[12, 31]$ for $n = 9$ (441 classes) and $[12, 36]$ for $n = 11$ (89,125 classes); the minimum 12 is attained by exactly two classes for $n = 7, 9, 11$, one of them the Siamese square. For $n = 7$ the other is the square $n\langle i + 2j - 2m\rangle + \langle i + j - m\rangle + 1$ obtained by exchanging the two digits of the closed form; it is magic for every odd n, and its gem has f-vector $(12, 24, 14)$ for $n = 5, 7, \ldots, 31$.

7 Open questions

1. *Hull versus point set.* Is $\mathrm{Iso}(\mathcal{G}(S)) = \mathrm{Iso}(P_S)$ for every magic, or semi-magic, square S? By Theorem 4.2 this would determine the isometry groups of all gems; it holds for order four (Theorem 4.3) and for the Siamese squares (Theorem 6.2). For odd n the self-complementary squares are associative (Lemma 4.1), so the point set of an order-five square has isometry group $\{\pm\mathrm{id}\}$ or trivial; whether the same holds for the gems is open.

2. *Order five.* Schroeppel's enumeration gives 275,305,224 essentially different squares of order five [13]. At the rate of the order-four computation (about 6 ms per exact hull) a full census would take about three CPU-weeks. How many combinatorial types and congruence classes occur, and are the congruence classes again the complement orbits?

3. *Growth of the vertex count.* For uniformly random arrangements, with 10^5 samples for each n, the mean vertex count is 8.22, 12.06, 15.39, 18.39, 21.08, 23.53, 25.79, 27.91 for $n = 3, \ldots, 10$ (standard errors at most 0.01), and the ratio $\mathbb{E}[V]/\ln^2 N$ decreases slowly from 1.70 to 1.32. For N independent uniform points in a 3-polytope the expected number of vertices of their hull is $\Theta(\ln^2 N)$ [3]. Is $\mathbb{E}[V] = \Theta(\ln^2 n)$ for random arrangements, and for uniformly random magic squares? Samples of magic squares obtained by simulated annealing (`anc/sa_magic.c`; 59,993, 24,000, 8,000 and 2,400 squares for $n = 5, \ldots, 8$) have mean vertex counts 15.38, 18.41, 21.08, 23.56, within 1.2 standard errors of the random values; the sampling distribution of the annealer is not known (compare the Monte Carlo methods of [24]). At $n = 4$ the exact magic mean is $269/22 \approx 12.23$.

4. *Secondary cones.* Which pairs of secondary cones of the $n \times n$ grid contain $(z, -z)$ for the height function z of a magic square (Proposition 2.5)? For $n = 4$ the 7040 magic height functions induce 417 regular subdivisions up to D_4.

5. *Linear squares.* Which linear squares (Remark 6.3) have a vertex count bounded independently of n, and does the digit-exchanged Siamese square of Remark 6.3 have f-vector $(12, 24, 14)$ for every odd $n \geq 5$?

6. *Other families.* The construction applies to semi-magic, pandiagonal and Franklin squares [17], and Proposition 2.3 and Theorem 4.2 already cover the semi-magic case.

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A Computational methods

All computations use Python 3.9 with NumPy 2.0, SciPy 1.13 (Quickhull), SymPy 1.13 and NetworkX 3.2; the scripts and data are the ancillary files in `anc/`, whose `README` lists the commands.

Exact hulls. Doubling all coordinates makes every point an integer vector. A floating-point Quickhull triangulation is used only to propose candidate facet planes, which are recomputed as primitive integer equations from three points each; coplanar triangles are merged. The hull is then certified in integer arithmetic: every one of the N points satisfies every facet inequality, every edge of every facet polygon lies in exactly two facets (so the facets form a closed surface and are all the facets), and $V - E + F = 2$. Vertices are the points on at least three facet planes, interior points are those satisfying every facet inequality strictly, and volumes are exact rationals. For the 880 order-four gems this takes about 6 s in total.

Combinatorial types, isometries, congruence. Combinatorial types are the isomorphism classes of the bipartite vertex–facet incidence graphs, grouped by a Weisfeiler–Lehman hash and confirmed by explicit isomorphism tests. Isometry groups of a finite set X (a vertex set or a full point set) are computed exactly: X is centred at its centroid, every isometry is then orthogonal and determined by the images of three linearly independent points, which are searched among the points with the same squared norms and mutual inner products, and a candidate is accepted only if the induced permutation of X preserves all pairwise integer squared distances. Two finite sets are congruent if and only if there is a bijection between them preserving all pairwise distances; this is tested by grouping by the sorted multiset of squared distances and then by an isomorphism test of the complete graphs with squared distances as edge labels. The point-set statements of Theorem 4.2 are checked in the same way for all 880 squares and for the Siamese squares of orders 3 to 15.

Polynomial identities. The slack polynomials of Table 3 are computed with SymPy; a polynomial p is certified positive for all integers $m \geq 2$ by non-negativity of all coefficients of $p(2+t)$ and $p(2) > 0$. The same script checks the closure of the facet list, the convexity of the quadrilaterals for $m \leq 60$, and the norms and inner products of Step 5.

Sampling. The random-arrangement statistics of Section 7 use Qhull in floating point, with the vertex sets of the first 100 hulls of each batch rechecked in exact arithmetic; the magic squares for $n = 5, \ldots, 8$ were generated by simulated annealing on the sum of squared line deviations (`anc/sa_magic.c`).